Shor's algorithm: the essentials

The story in one sentence. Shor's algorithm breaks RSA and elliptic-curve keys by finding a hidden period with a quantum computer. Almost all of its cost is ordinary arithmetic made reversible, and the 2026 work, including this repo, is about making that arithmetic cheap.

1 · How Shor works (Part 1)

- **Factoring is period finding.** The powers of 7 mod 15 run 1, 7, 4, 13, 1, … so the period is r = 4. Then gcd($7^2 - 1$, 15) = 3 and gcd($7^2 + 1$, 15) = 5. Finding r is the only quantum step.
- **The circuit (phase estimation).** A counting register in superposition controls "multiply by a^(2^j) mod N" on a work register; then an inverse QFT, then one measurement y.
- **Phase kickback.** The work register is made of eigenvectors of "multiply by a", so each controlled multiplication only adds a phase, and that phase lands on the control. The counting register ends up "spinning" at frequency s/r: a phase ramp.
- **The inverse QFT is a frequency meter.** It turns the ramp into a peak at y ≈ Q·s/r; continued fractions turn y/Q into r.
- *Say:* "One fixed circuit pattern; only the box inside it changes."

2 · What it costs (Part 2)

- **Error correction.** Real qubits fail about once per 1,000 operations. The surface code stores one *logical* qubit in about 1,500 physical ones (a grid, distance d ≈ 27): parity checks catch errors without reading the data.
- **The cost unit is the T gate.** On error-corrected hardware H, S and CNOT are cheap; T needs a "magic state" from a factory. A **Toffoli** (reversible AND) costs 7 T; a small rotation costs 60–100 T, about ten Toffolis. So arithmetic is built from Toffolis.
- **Garbage must be cleaned up.** Leftover scratch that depends on x acts like a which-path detector and destroys the interference, so every scratch qubit is uncomputed: compute, copy, uncompute, about 2× the cost.
- **RSA-2048:** the textbook circuit needs 6.9×10^{10} Toffolis; Gidney 2025 needs 6.5×10^{9} on 1,399 logical qubits, under a million physical qubits for about five days.

3 · How the cost was cut (Part 3): rotations first, then qubits, then additions

- **Ripple-carry adders** (no rotations): T −95 to −99%. **One counting qubit**, measured and reused: qubits −41 to −50%.
- **Temporary AND:** 4 T to compute, free to undo. **Windowing:** look up w bits at once, so n additions become n/w. **Measured uncomputation:** undoing a lookup costs −96%.
- **The key trick:** measuring a scratch qubit in the X basis reveals nothing about the data; it only leaves a known sign, which is fixed cheaply.
- **N = 15, all together:** 978,000 T → 42,600 T (−96%).

4 · Elliptic-curve keys (Part 4)

- **The key is a 2-D period.** Secret k, public Q = [k]P. Then f(u, v) = [u]P + [v]Q = [u + kv]P repeats when (u, v) moves by (−k, 1): the hidden subgroup L = ⟨(−k, 1)⟩.
- **Read-out.** Each run gives (s, t) with t ≡ k·s (mod r), so **$\mathbf{k = t \cdot s^{-1} \bmod r}$**. The order r is public, so there are no continued fractions.
- **The box adds points**, and a point addition needs a modular **division**, done twice (compute, then uncompute): 80% of the cost.
- **Why curves fall first:** cost grows as key length^3, and at the same security the key is 256 bits, not 3,072. $12^3 \approx 1{,}700$, divided by a ~7× heavier operation: **150–270× fewer Toffolis**.
- **So:** recorded traffic can be decrypted later; migrate to post-quantum ML-KEM and ML-DSA.

5 · Where things stand (Part 5)

- **ECDLP-256 in 2026:** about 1,200–1,500 logical qubits and $4\text{–}9\times10^{7}$ Toffolis. Google's design: under 500,000 superconducting qubits for about 20 minutes; ions or atoms: about 20,000 qubits for days to weeks. No such machine exists yet.
- **This repo** built and counted every circuit: **1.31M Toffolis per point addition on 1,457 qubits, 5.9% below IonQ's published count**; the whole ECDLP-256 is 37.1M Toffolis, −83% against the papers as printed.

Questions, short answers

- **"When will Bitcoin break?"** No date: these are estimates for machines that don't exist, and the cheapest designs rest on the least-tested assumptions. Migrate anyway.
- **"Doesn't measuring destroy the interference?"** Only a measurement that reveals x. An X-basis measurement reveals a coin flip and leaves a known sign that is fixed.
- **"Is f(u, v) periodic?"** Yes, with a lattice of periods; (−k, 1) is the one that hides the key.
- **"Are these numbers yours?"** The repo's circuits are built, counted and checked on every input at toy sizes; the papers' numbers are quoted as reported.
- **Don't know?** "Good question — the full deck has a slide on it; I'll follow up."

On the day

- **Running late?** Skip every slide marked "cut first". Checkpoints: slide 39 at about 24 min, S2 at 34, slide 109 at 46, slide 211 at 55.
- **On each slide:** point first, then talk; say one number per slide; read tables in ten seconds, then stop on two rows.